

## **SUPPRESSION OF CELL PROLIFERATION BY SNAKE VENOM COULD BE RELATED WITH A FALL OF NUCLEIC ACID CONTENT IN DMBA – INDUCED RABBIT SKIN CARCINOMA**

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### **ABSTRACT:**

Venom has been reported to suppress the cell proliferation, a process which leads to the genesis of cancer. An attempt has been made to observe the *in vitro* effect of different concentrations (10, 25, 50 µg/ml) of crude cobra snake venom (ccsv), on the contents of RNA and DNA of normal and cancerous (DMBA- induced skin cancer) tissues.

The venom failed to demonstrate any change in nucleic acid content of the normal (control) rabbit skin as against DMBA-affected skin. A significant fall in nucleic acid concentration, more pronounced with 25 µg/ml dose, was evident.

The data thus suggest that crude snake venom do have a direct suppressant effect on nuclear processes which lead to cell proliferation.

### **INTRODUCTION**

(Calman *et al.*, 1980 and Maylow, 1993).

In early seventies (70's), the therapeutic use of individual components of snake venom in a variety of diseases including cancer, attracted the attention of proper scientific personnel (Jimenez-Porras, 1970; Cotias *et al.*, 1975; Calabresi and Parks, 1976; Braganca, 1976 and Harvey, 1995). However, owing to controversy in the reports, this beneficial effect could not found proper meritorious place in the therapeutics.

The treatment of cancer has evolved significantly over the years in order to achieve the therapy with minimal morbidity and the highest quality of life (Pratt *et al.*, 1994). An initial step towards the cure of cancer by snake venom and other toxins was claimed by some physicians in early 20<sup>th</sup> century (Grunbaum, 1911 and Menegbetti, 1935). Recently, chemotherapy of cancer has become a promising approach (Pratt, 1994 and Ashmawy, 1999), because surgery and radiation have been found to be not fully effective against the high incidence or low survival rate for man metastatic tumours

Like United States of America and several European countries, many Asian populations also have been affected by many cancers (Calabresi, 1990; Lindholm, 1993 and Maylow, 1993). In Pakistan out of 140 million people, approximately 200,00 new cases of cancer are diagnosed every year. According to general cancer surveys, there has been a 40% increase in cancer patients over the last six years (Pashah, 1993) and this seems to be alarming.

In the present study, cancer was induced on rabbits skin and the cancerous tissue was studied to observe the suppressant effect of cobra venom on the cell proliferation through determinations of nucleic acids. Crude cobra venom has been well documented as growth inhibitor and even repressor of tumours following direct local administration into neoplasm (Dustin, 1939; Hode and Thorn, 1941). This also has been shown to possess cytotoxic or lytic effects in *in vitro* studies carried out on the tumour cells (Tu, 1991; Newman, 1993 and Harvey, 1995). Thus,

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snake venom might prove a useful weapon of the future, to combat the cancer disease.

## MATERIALS AND METHODS

### *Materials:*

#### **1. Chemicals required**

Nucleic acids (RNA and DNA), Dimethyl benz- anthracene (DMBA) were supplied by

Fluka, for standardization. All other reagents were of 'Analar grade' and were supplied by BDH Chemicals, Ltd., Poole (England) and E. Merck (USA) and Riedel-de Haen AG Seclze – Hannover (Germany).

#### **2. Sample collection:**

The experiments were performed on rabbit skin (normal and cancerous) tissues. The animals were supplied by Alberuni

Fig 1: DMBA-Induced skin cancer in rabbit

Scientific Stores, Hyderabad and kept in the departmental animal house to adapt natural environment. The rabbits were randomly divided into two groups, out of which one served as control and the other was given alternate applications of DMBA (a carcinogenic) and croton oil (cancer promoter) on selected areas, to produce skin cancer as shown in Fig.1 (photograph).

#### Animal Tissue Collection:

Skin tissues (Rabbit /1100—1200 gms) were taken out from normal and DMBA – affected animals according to the method of Ceriotti (1955). That is; skin tissues of rabbits anaesthetized locally with xylocaine (2%) were obtained in the beaker containing ice-cold normal saline. After three washings with saline, the tissues were pressed within the filter paper and weighed.

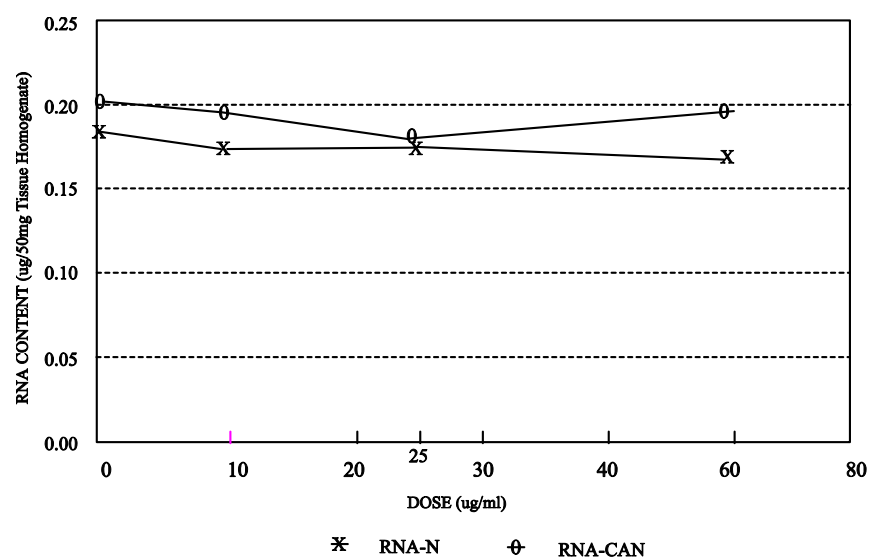


Fig. 2: Amount of RNA in venom treated skin tissues of normal and cancer (rabbit).

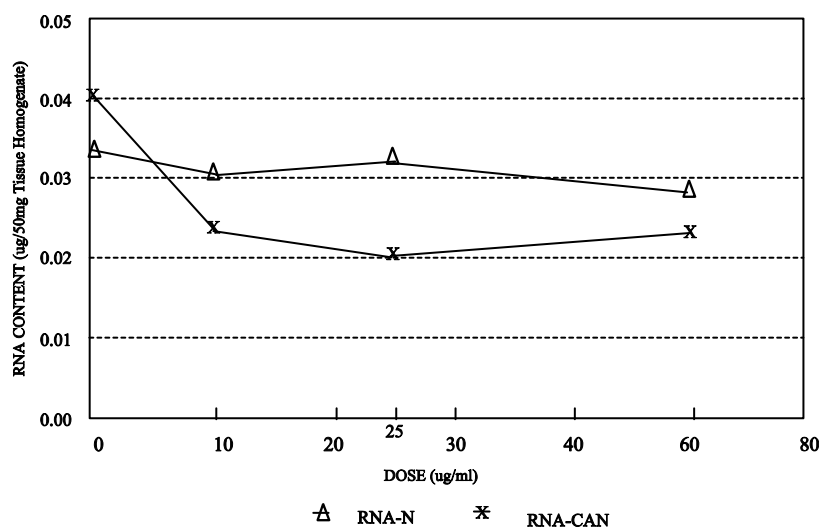


Fig. 3: Amount of DNA in venom treated skin tissues of normal and cancer (rabbit).

### 3. Homogenate preparation, incubation and extraction:

Tissues were cut into small pieces with the help of a sharp sterilized razor and put into the homogenizer (k-ika-werk-Ultra-Turrax). The homogenates were made with saline (1:3 w/v) as reported by Ceriotti (1955). These homogenates were then diluted and treated in known volume of ice cold saline. All tissues were kept in a deep freezer till further experimentation. Before incubation and extraction, all the tissue aliquotes of each homogenized solution were treated with various concentrations (10, 25 and 50 µg/ml) of the venom, in triplicate, for 30 min at 37°C,

followed by acidification and extraction in chloroform.

### 4. Determinations:

- DNA – content determinations were carried out in triplicate by the modified indole method of Ceriotti (1955).
- RNA – content determinations were carried out in triplicate by the Orcinol-method of Schneider (1957). Bauch and Lamb, spectrophotometer, Spectran-21 was used to measure DNA at 490 nm and RNA at 660 nm. Results for control and experimental groups were compared using students 't' test.

**Table 1(a)**

Amount of total Nucleic Acid in Normal Rabbit Skin Tissues incubated with and without Cobra Venom (25 µg/ml)

| Tissues             | Nucleic Acid    |                 | Inhibition whether significant or not significant |
|---------------------|-----------------|-----------------|---------------------------------------------------|
|                     | RNA (µg/50ml)   | DNA (µg/50ml)   |                                                   |
| Normal              | C 0.184 ± 0.001 | C 0.033 ± 0.001 | N.S.                                              |
| Rabbit skin tissues | S 0.152 ± 0.015 | S 0.031 ± 0.001 | N.S.                                              |

C = Control (without snake venom); S = Sample (treated with snake venom); (Average ± SEM)

**Table 1(b)**

Nucleic Acid Levels in Normal Tissues of Rabbit Skin Treated with varying concentrations (10, 25, 50 µg/50ml) of Crude Cobra Venom

| Tissues             | Nucleic Acid                 |                              | Inhibition whether significant or not significant |
|---------------------|------------------------------|------------------------------|---------------------------------------------------|
|                     | RNA (µg/50ml)                | DNA (µg/50ml)                |                                                   |
| Normal              | C 0.184 ± 0.001              | C 0.033 ± 0.001              | S                                                 |
|                     | S <sup>1</sup> 0.150 ± 0.016 | S <sup>1</sup> 0.030 ± 0.001 | N.S.                                              |
| Rabbit skin tissues | S <sup>2</sup> 0.152 ± 0.015 | S <sup>2</sup> 0.031 ± 0.001 | N.S.                                              |
|                     | S <sup>3</sup> 0.153 ± 0.015 | S <sup>3</sup> 0.028 ± 0.001 | N.S.                                              |

C = Control (without snake venom)

S<sup>2</sup> = Sample with 25µg/ml snake venom  
(Average ± SEM)

S<sup>1</sup> = Sample with 10µg/ml snake venom

S<sup>3</sup> = Sample with 50µg/ml snake venom

S = P<

## RESULTS

In this study, the *in vitro* effect of crude cobra snake venom (ccsv) on different cellular macromolecular (nucleic acids; RNA and DNA) has been quantitatively assessed using normal and cancerous animal tissues. An attempt has also been made to assess the effect of different concentrations (10, 25, 50 µg/ml) of the venom, after skin cancer was experimentally induced in rabbits at our labs.

Cancer was induced on to the body of a rabbit (as was proposed by Gijare, 1989) using 9, 10 – dimethyl – 12-benz-anthracene (DMBA) as carcinogen and croton oil as promoting agent. After 3 months treatment cancer was visible on the skin of the rabbits as shown in Fig.1. The effect of snake venom treatment on newly induced cancer tissue and on the normal tissue was studied by measuring their nucleic acid contents before and after incubation with snake venom. The estimated levels of these macromoles are given in Table-1, both the tissues were also subjected to varying concentrations, (10, 25, 50 µg/ml) of the venom and the data thus obtained has been presented in the form of curves as shown in Figs. 2 and 3.

From the data, it is evident that no significant change in the amounts of RNA and

DNA has occurred in the normal skin sample whereas, cancerous tissue (chemically-induced) sample which have large concentrations of nucleic acids at basal level, showed a significant fall in the RNA and DNA contents.

The corresponding values regarding the reduction of RNA/DNA-levels indicate that cancerous nucleic acids are more sensitive towards the venom. The dose response curve shows that the venom concentration (25 µg/ml) produced maximum inhibition of both nucleic acids and conversely the increasing doses did not produce any significant change.

From this observation, it can be assumed that venom at 25 µg/ml dose might have maximum reduction or therapeutic effect in cancerous tissues.

## DISCUSSION

The present study was designed to investigate whether venom does have any effect to alter cellular nucleic acid levels and if it had any; then to compare its effect in the normal as well as cancerous animal tissues within the range of the dose regime being studied throughout.

The yield of tumours was substantially

**Table 2**  
The overall results showing the effects of various concentrations of cobra snake venom in the RNA and DNA (Average ± S.E.M.) Levels of rabbit skin (DMBA-induced) Tissues, Summarized

| Tissues                | Nucleic Acid (RNA, DNA) | Control (Normal & Cancerous) | Various Concentrations of Venom |               |                | Inhibition whether significant or not significant |
|------------------------|-------------------------|------------------------------|---------------------------------|---------------|----------------|---------------------------------------------------|
|                        |                         |                              | I<br>10µg/ml                    | II<br>25µg/ml | III<br>50µg/ml |                                                   |
| Normal and skin cancer | RNA (N)                 | 0.184 ± 0.001                | 0.150 ± 0.016                   | 0.152 ± 0.015 | 0.153 ± 0.015  | N.S.                                              |
|                        | RAN (Ca)                | 0.207 ± 0.03                 | 0.193 ± 0.002                   | 0.179 ± 0.003 | 0.198 ± 0.002  | S                                                 |
| Tissues                | DNA (N)                 | 0.033 ± 0.001                | 0.030 ± 0.001                   | 0.031 ± 0.001 | 0.028 ± 0.001  | N.S.                                              |
|                        | DNA (Ca)                | 0.040 ± 0.001                | 0.024 ± 0.001                   | 0.020 ± 0.001 | 0.024 ± 0.001  | S                                                 |

N = Normal; Ca = Cancerous; Control = Without snake venom; (Average ± SEM)

increased when the carcinogen, DMBA was repeatedly applied over a period of three months, not only that tumour was raised considerably by the simultaneous application of a non-carcinogenic substance croton oil (as a promoter). The newly induced cancerous tissues were then studied for the effects of different doses of cobra snake venom.

The data obtained showed that cancer tissues (DMBA-induced cancer of rabbit skin) possessed considerably larger amounts of nucleic acids than did the normal tissues. This agrees with the observation of Brachet and Jeener, 1944; for the RNA/DNA that rapidly growing tissues always contain larger amounts of non-sedimentable DNA than differentiated tissues. (Seehar, 1984).

Since a necessary pre-requisite for cell division is the duplication of DNA. Therefore, in the control of mitosis, the initiation of DNA synthesis is a key event in the cell cycle (Killander and Zetterberg, 1965). And RNA's increased content (cancer tissue) has been correlated with endocrine activity of thyroid and the adrenal cortex (Motavino, 1951) as well as anterior pituitary (Hess, 1959). Whereas, increased DNA content indicates the occurrence of cell proliferation, although, DNA is essentially constant in amount per nucleus in non-proliferating cells (Naora, 1957). Thus increase in the amount of mitotic activity would be reflected by an increase in total DNA of an organ and should indicate an increment in the cell number (Danes, 1961). Increased RNA content in cancer tissues could be due to increased DNA-dependent RNA polymerase.

This study has been carried out for assessing the effect of venom on the proliferating cells by measuring the levels of macromolecules that are responsible for cellular growth. The inhibitory role of venom in various physiological processes including transport mechanism of cell membrane and cell's mitotic division has been well documented (Griffin, 1948; Price *et al.*, 1949 and Braganca, 1978).

Some authors reported to have observed, the growth inhibitions and even regressions of tumors in animals after the direct administration of cobra venom into neoplasm (Hode and Thron, 1941 and Calmette, 1933). In 1939 special attention was called by Dustin and his collaborators for *in vivo* and *in vitro* studies to see the influence of toxic substances on cell division and growth. These results led to the suggestion that snake venom might be helpful in the treatment of cancer.

Our results clearly show a suppression in DNA, and RNA contents when the cancerous tissues were treated with cobra venom. Various studies reported in the literature have demonstrated that cobra venom components like phospholipase, cardiotoxin and many other enzymes especially endonucleolytics (DNases and RNases) do inhibit cancer growth (Gillo, 1966; Meldrum, 1965 and Holley, 1968).

We observed no significant change in the concentration of nucleic acids when normal tissues were treated with varying concentrations of the venom. In contrast, the cancerous tissues responded well and the skin (affected) samples got their DNA and RNA contents reduced to maximum again with 25 µg/rnl of venom. This strengthens our previous findings of venom's effect on cellular electrolytes fluxes and sodium-pump activity (Ansari *et al.*, 1989) and hence it could be proposed that the said dose of venom could be more practicable.

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